OCEANOGRAPHIC CONDITIONS AND SMALL PELAGIC FISHERY DYNAMICS ALONG THE WEST COAST OF BAJA CALIFORNIA IN 2025

Created by:

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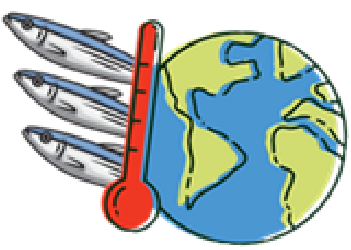

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ABSTRACT

This technical report assesses the oceanographic conditions during 2025 and their relationship with small pelagic fishery catches along the West Coast of the Baja California Peninsula (WCBCP, designated as Region A). Region A is an integral component of the highly dynamic California Current System (CCS). The study integrated multiple environmental variables including sea surface temperature (SST), chlorophyll-*a* concentration (CHL-*a*), sea level anomaly (SLA), geostrophic currents, and Bakun upwelling indices alongside landing statistics from CONAPESCA (2026) to evaluate spatio-temporal resource availability and support adaptive fisheries management. The physical oceanography of the region exhibited clear seasonal and latitudinal transitions during 2025. Winter and early spring (January to April) were defined by cool baseline SSTs (14–18 °C), widespread negative SLAs (−20 to 0 cm), and the onset of wind-driven coastal upwelling. These processes triggered extensive primary productivity, with CHL-*a* concentrations exceeding 4 mg/m^3 off key coastal upwelling zones. A transitional warming period in May and June gave way to the summer–autumn peak (July to October), during which a strong latitudinal thermal gradient established. Tropical water intrusion raised southern SSTs to 24–26 °C, while northern subarctic waters remained between 18 °C and 21 °C. Concurrently, upwelling collapsed in the south, primary productivity dropped to background levels (0.05–1 mg/m^3), and positive SLAs (30 to >40 cm) accompanied strong northward geostrophic flows. Fishery landings were heavily dominated by the Pacific sardine, which comprised over 50% of catches in most months and reached 80–100% from September to December. Northern anchovy represented the second largest capture volume, peaking during spring and early summer (up to 40–50% of monthly catches).

INTRODUCTION

This report describes the oceanographic conditions during 2025 and the catches of small pelagics along the West Coast of the Baja California Peninsula (WCBCP), designated as Region A for the exploitation of small pelagics (DOF, 2019). This zone forms part of the California Current System (CCS), a dynamic ecosystem that experiences seasonal variations in its physical characteristics and flow patterns, driven primarily by interactions between currents, eddies, and large-scale oceanographic processes (Lynn & Simpson, 1987).

According to Durazo (2015), the hydrographic conditions off the Baja California peninsula indicate that the region is homogeneous during winter and spring, characterized by low temperatures and low salinity, which imparts a subarctic character. In contrast, during summer and autumn, it subdivides into two distinct regions: the area north of Punta Eugenia (28°N) retains subarctic features, whereas the southern region receives an influx of tropical and subtropical waters. The distinct characteristics of the water masses transported by the California Current and the California Undercurrent determine their properties along their trajectory.

These ocean-atmosphere interactions not only define the thermal structure of the water column but also drive primary productivity within this biogeographical transition zone (Lluch-Belda, 2000). Within this physical framework, small pelagics have developed highly specialized life-history strategies, with their distribution and reproductive success serving as a direct reflection of environmental fluctuations (Parrish & Mallicoate, 1995; Valencia-Gasti et al., 2015).

Key species such as the Pacific sardine (*Sardinops sagax*), northern anchovy (*Engraulis mordax*), and Pacific mackerel (*Scomber japonicus*) exhibit pronounced sensitivity to habitat variations (Baumgartner et al., 1992). This vulnerability becomes particularly evident during warm events such as the El Niño–Southern Oscillation (ENSO), where increases in sea surface temperature and reductions in nutrient availability lead to drastic declines in catch volumes. A documented historical case occurred during the intense 1992 event, when *S. sagax* landings in Magdalena Bay collapsed from 16,000 t in 1991 to fewer than 5,000 t within a single year (Félix-

Uraga, 1996). Conversely, cold episodes generally establish optimal environmental conditions that foster reproductive success and boost the abundance of these small pelagic stocks (Parrish & Mallicoate, 1995).

In this context, systematic monitoring of the WCBCP is essential to understand the interplay between climate variability and local biological dynamics, as well as being fundamental to anticipating changes in resource availability. Therefore, this bulletin aims to integrate sea surface temperature (SST), chlorophyll-*a* concentration (CHL-*a*), sea level anomaly (SLA), and upwelling indices to deliver a technical assessment that supports adaptive management of small pelagic fisheries in the region.

STUDY AREA

The WCBCP is defined as one of the three small pelagic fishing regions under NOM-003-SAG/PESC-2018. Region A encompasses the distribution zone of small pelagics along the west coast of the Baja California peninsula. It is bounded to the north by the border with the United States of America and to the south by an imaginary line passing through the 23°00' North parallel (DOF, 2019).

To ensure appropriate monitoring of fishing activity, 9 areas based on the polygons used by the California Cooperative Oceanic Fisheries Investigations (CalCOFI) program and 28 subareas were established within Region A (Figure 1).

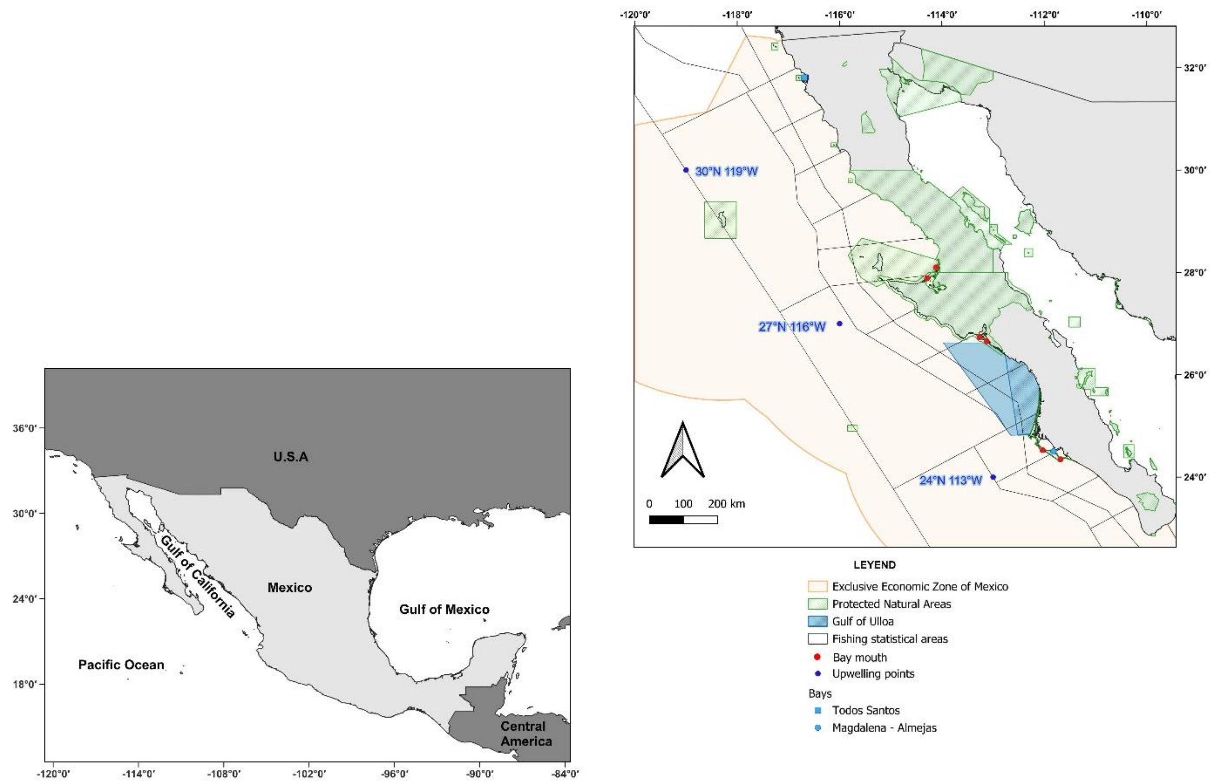

Figure 1: Spatial distribution of fishing statistical areas off the west coast of the Baja California Peninsula (Region A). The map highlights Protected Natural Areas (PNAs), the Gulf of Ulloa, Mexico's Exclusive Economic Zone (EEZ), key bay mouths, upwelling sampling points and major bay systems (Todos Santos and Magdalena-Almejas).

METHOD

Oceanographic conditions off the west coast of the Baja California peninsula (WCBCP)

Sea surface temperature

Sea surface temperature data for the year 2025 were obtained from EARTHDATA, derived from NASA's Moderate Resolution Imaging Spectroradiometer (MODIS) sensors, with a spatial resolution of 1 km and a temporal resolution of 1 day (JPL MUR MEaSUREs Project, 2015). The downloaded satellite imagery was processed in R version 3.2, where the monthly averages were calculated and values were converted from Kelvin to degrees Celsius. Subsequently, each weekly image was exported in NetCDF format for map generation in QGIS version 3.40.6.

Chlorophyll-*a*

Chlorophyll-*a* data for 2025 were obtained from OCEAN COLOR, derived from NASA's Moderate Resolution Imaging Spectroradiometer (MODIS) sensors, with a spatial resolution of 4 km and a temporal resolution of one month (NASA Goddard Space Flight Center, Ocean Ecology Laboratory, Ocean Biology Processing Group, 2014). The downloaded satellite imagery was processed in QGIS version 3.40.6 to generate monthly maps.

Altimetry

This analysis was conducted using data from the Copernicus Marine Service for product SEALEVEL GLO PHY L4 NRT 008 046 (Copernicus Marine Service, n.d.), with a spatial resolution of 13 km and a temporal resolution of 1 day. Given the relatively low spatial resolution and that the data stem from an interpolation process, results should be interpreted with caution, particularly near the coast. In the Gulf of California, sea level anomalies exceed 30 cm (the maximum value recorded for the Pacific), leading to missing data in some areas, represented in white.

The variables extracted from the Copernicus Marine Service to derive altimetry were sea surface height above sea level (sea level anomaly; SLA), surface geostrophic eastward sea water velocity assuming sea level for geoids (UGOSA), and surface geostrophic northward sea water velocity assuming sea level for geoids (VGOSA). The downloaded satellite imagery was processed in R version 3.2, where monthly averages were calculated. Subsequently, each monthly image was rendered as a map.

Upwelling Indices

Bakun upwelling index data were obtained at three points along the WCBCP: 30°N, 119°W, off Punta Baja; 27°N, 116°W, southwest of Punta Eugenia; and 24°N, 113°W, off Magdalena–Almejas Bay (Figure 1; Environmental Research Data Services, n.d.). These indices are derived from Ekman's theory, which describes surface water transport induced by wind stress. According to this theory, under homogeneous, uniform wind, and steady-state conditions, transport occurs at approximately 90° to the right of the wind direction in the Northern Hemisphere (Bakun, 1973). This displacement of surface water prompts the upwelling of deep, nutrient-rich waters, a process that characterizes coastal upwelling. Thus, positive Bakun index values reflect active upwelling conditions, whereas negative values indicate downwelling or subsidence, where surface water accumulates and deep-water upwelling does not occur (Bakun, 1973).

Time series of Todos Santos Bay and Magdalena–Almejas Bay

To characterize interannual variability and sea surface temperature anomalies, a time series spanning from January 2023 to January 2026 was generated. Processing was based on the satellite products described above, focusing the analysis on two key regions off the West Coast of the Baja California Peninsula: Todos Santos Bay and Magdalena–Almejas Bay variable values were standardized, where Z = 0 represents conditions identical to the historical average of the series. Positive values denote above-average conditions (positive anomaly), whereas negative values indicate below-average conditions (negative anomaly), thereby facilitating the visual identification of extreme events.

Climate events

To contextualize local SST variations within a broader climatic framework, a comparative analysis was conducted against El Niño–Southern Oscillation (ENSO) and Pacific Decadal Oscillation (PDO) patterns. This procedure relied on monthly averages of the Southern Oscillation Index (SOI) and the Pacific Decadal Oscillation Index (PDOI), which enable the identification of anomalous phases in atmospheric pressure and North Pacific temperatures, respectively. Data corresponding to these indices were obtained from the National Oceanic and Atmospheric Administration's Climate Prediction Centre (NOAA, 2009, 2025).

Spatio-temporal catch patterns of small pelagics in Region A

To estimate the production for Region A, landing and harvesting notices from CONAPESCA for the year 2025 were utilised, which provide information in tonnes for Pacific sardine (*Sardinops sagax*), northern anchovy (*Engraulis mordax*), Pacific mackerel (*Scomber japonicus*), thread herring (*Opisthonema* spp.), Northern anchoveta (*Cetengraulis mysticetus*), Pacific thread herring (*Etrumeus teres*), and Pacific jack mackerel (*Trachurus symmetricus*).

RESULTS

Oceanographic conditions off the west coast of the Baja California peninsula (WCBCP)

Sea surface temperature

The monthly maps depict the spatio-temporal evolution of SST along the Baja California peninsula and within the Gulf of California, illustrating a marked seasonal cycle.

From January to April, cool conditions prevail across the region. Off the western coast of the peninsula, SSTs generally range between 14 °C and 18 °C, with the 17 °C isotherm extending southward past 27°N during late winter. Concurrently, the Gulf of California exhibits cool to moderate thermal conditions, hovering around 16 °C to 20 °C (Figure 2).

A clear transitional warming period occurs during May and June. During May, the 17 °C isotherm retreats northward along the Pacific coast, giving way to widespread values above 18 °C. By June, thermal contrast intensifies as the Gulf of California warms rapidly, reaching temperatures above 26 °C, peaking near 28 °C, whereas the Pacific coast retains cooler waters influenced by upwelling and the California Current.

The summer and early autumn months (July to October) represent the warmest phase of the year. Peak SSTs occur between August and September, when the entire Gulf of California displays temperatures exceeding 30 °C. Along the Pacific coast, a pronounced latitudinal gradient develops: the southern portion receives warm tropical waters reaching 24 °C to 26 °C, delineated by the 22 °C isotherm, while subarctic coastal waters north of Punta Eugenia (28°N) remain noticeably cooler (18 °C to 21 °C) (Figure 3 and Figure 4).

Finally, November and December mark the autumn-winter transition, characterized by rapid cooling across both basins. By December, Gulf temperatures drop below 22 °C, and the 17 °C isotherm reappears off the Pacific coast, returning the region to its characteristic winter baseline (Figure 4).

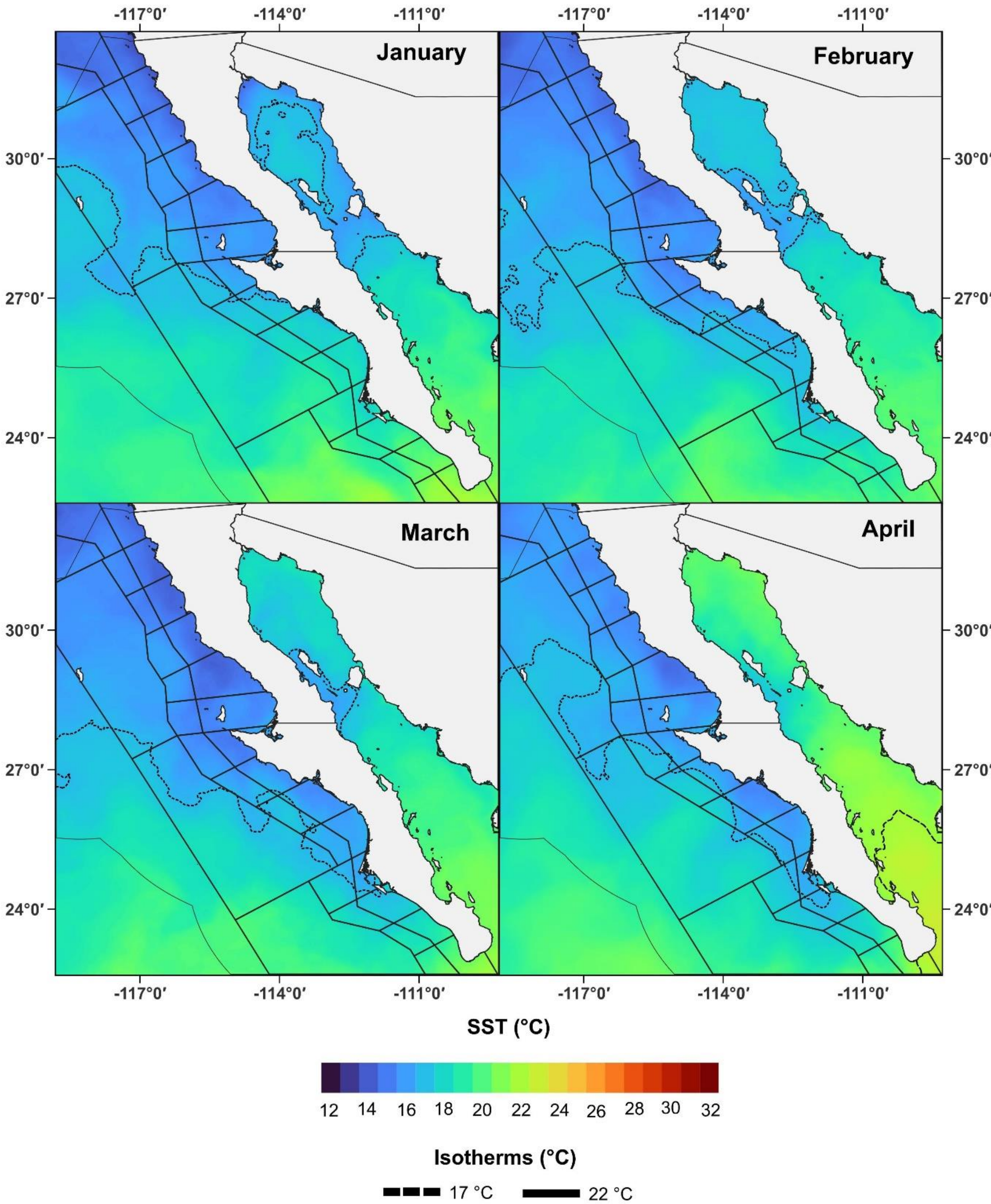

Figure 2: Satellite images of monthly SST averages from January to April 2025.

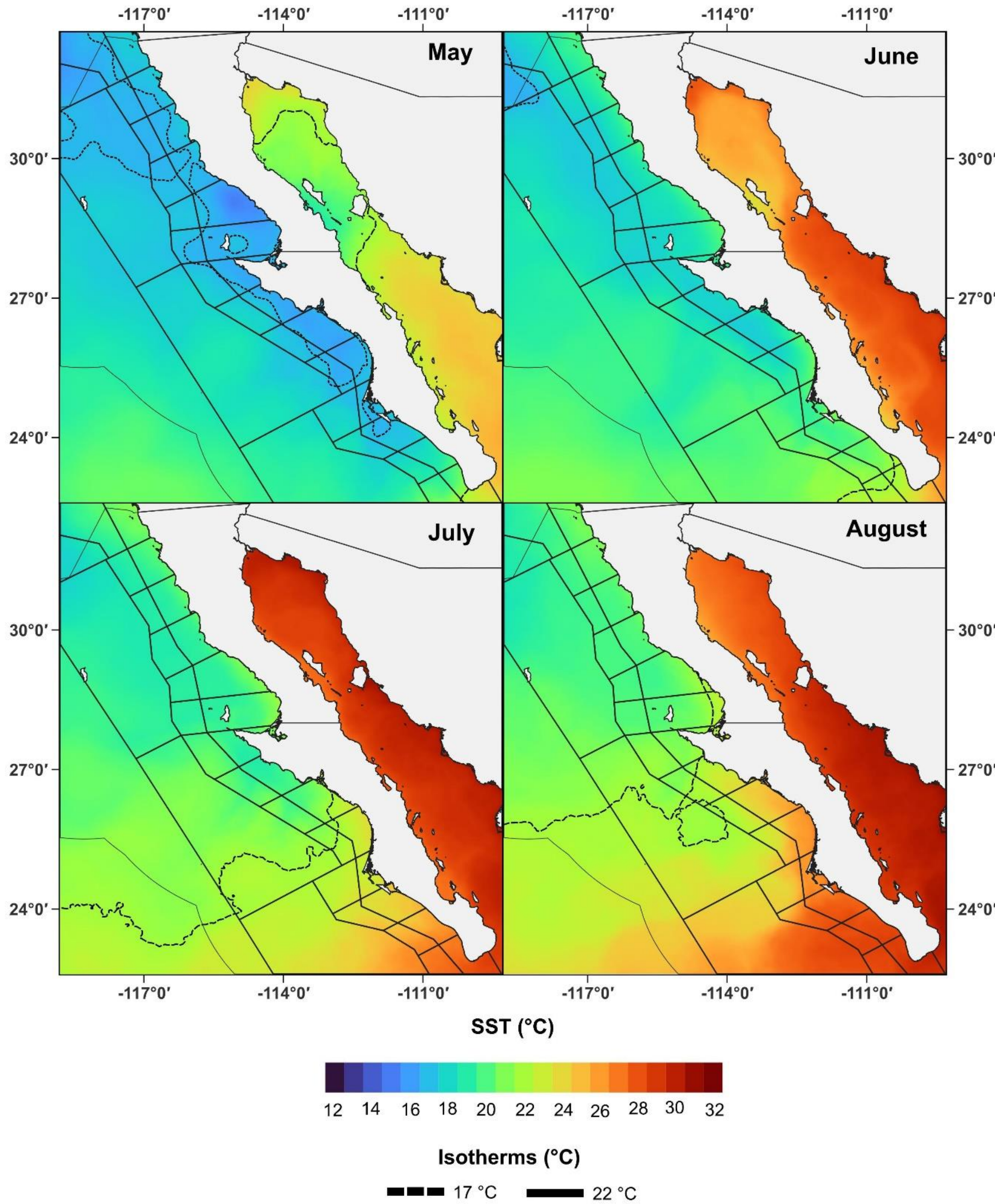

Figure 3: Satellite images of monthly SST averages from May to August 2025.

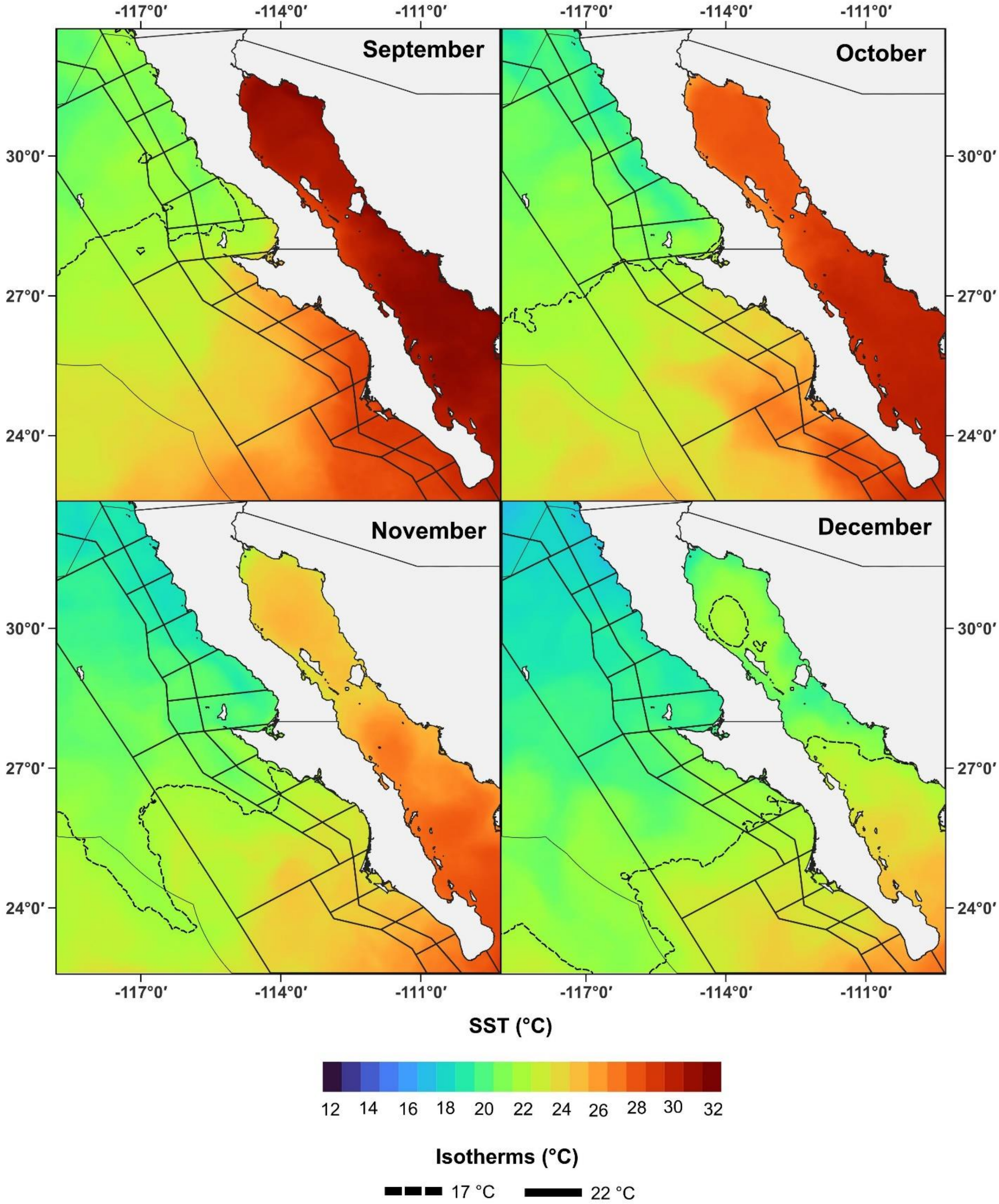

Figure 4: Satellite images of monthly SST averages from September to December 2025.

Chlorophyll-a

The monthly maps illustrate the spatio-temporal dynamics of satellite-derived chlorophyll-*a* (mg/m^3) across the Baja California Peninsula and the Gulf of California, highlighting clear seasonal patterns of primary productivity.

From January to April, a progressive intensification of primary production is observed. During winter, moderate chlorophyll-*a* levels (2-6 mg/m^3) emerge along the coast. By spring (March and April), wind-driven coastal upwelling triggers extensive phytoplankton blooms along the Pacific coast of the peninsula, particularly off Punta Baja, Punta Eugenia, and the Gulf of Ulloa, where concentrations frequently exceed 10 mg/m^3 near the shore. Simultaneously, elevated concentrations expand throughout the upper and central regions of the Gulf of California (Figure 5).

During late spring and early summer (May and June), high chlorophyll-*a* concentrations (8 mg/m^3) remain persistent along the Pacific coast, driven by strong seasonal upwelling. In contrast, productivity inside the Gulf of California begins to decline significantly, returning to oligotrophic conditions (<0.5 mg/m^3) across most of its central basin as thermal stratification sets in (Figure 6).

The late summer and early autumn period (July to September) represents the annual productivity minimum for the region. Coastal upwelling along the Pacific coast weakens substantially, restricting high chlorophyll-*a* concentrations to narrow inshore strips and shallow embayments such as Magdalena Bay. Throughout this period, both the offshore Pacific waters and the vast majority of the Gulf of California display low background levels (0.05-1 mg/m^3) (Figure 7).

In late autumn (October to December), primary production begins to recover. Chlorophyll-*a* concentrations rise notably along the eastern coast of the Gulf of California, driven by the onset of north-westerly winter winds, whilst low to moderate productivity persists along the Pacific coastline of the peninsula (Figure 7).

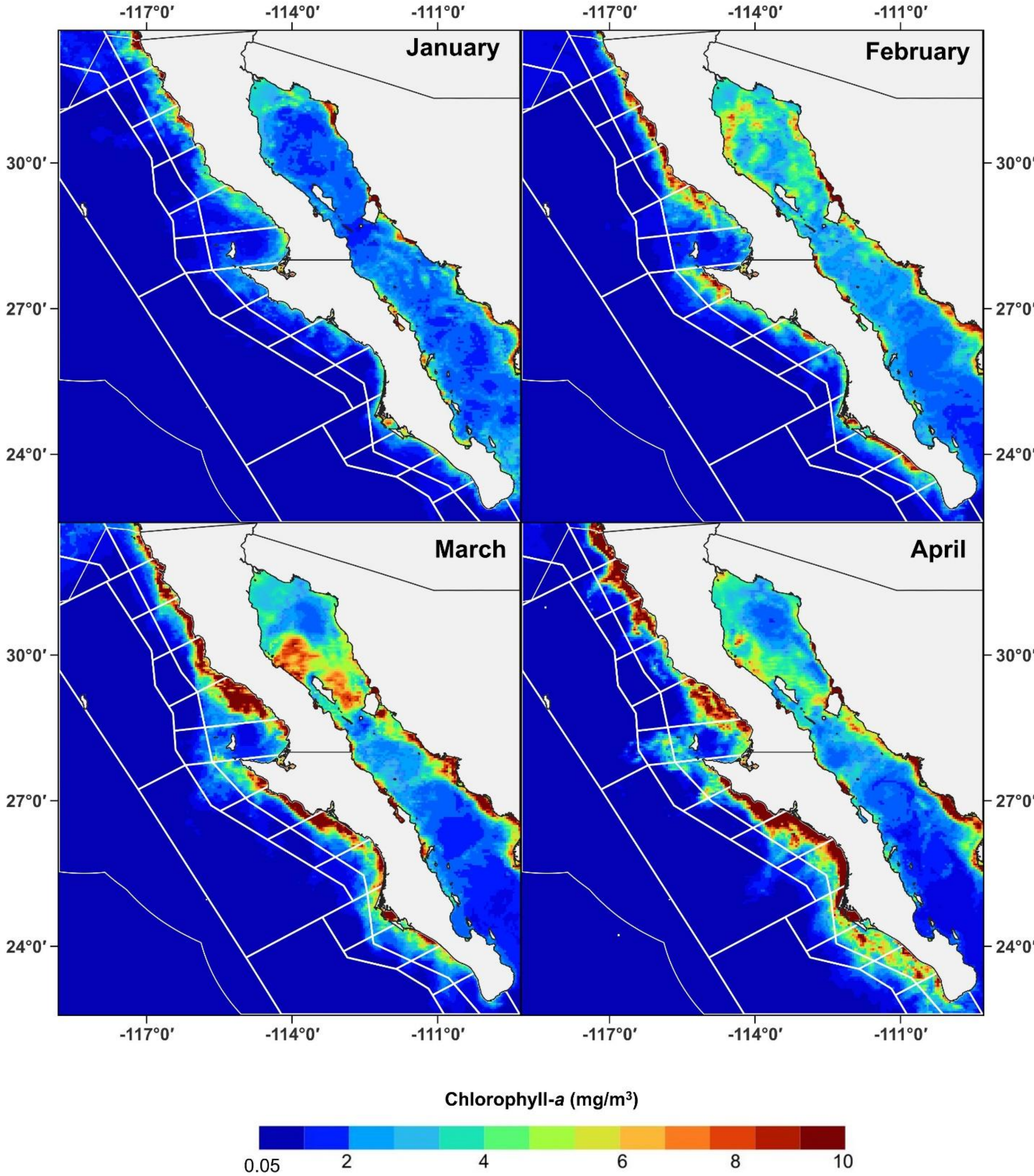

Figure 5: Satellite images of monthly CHL-a averages from January to April 2025.

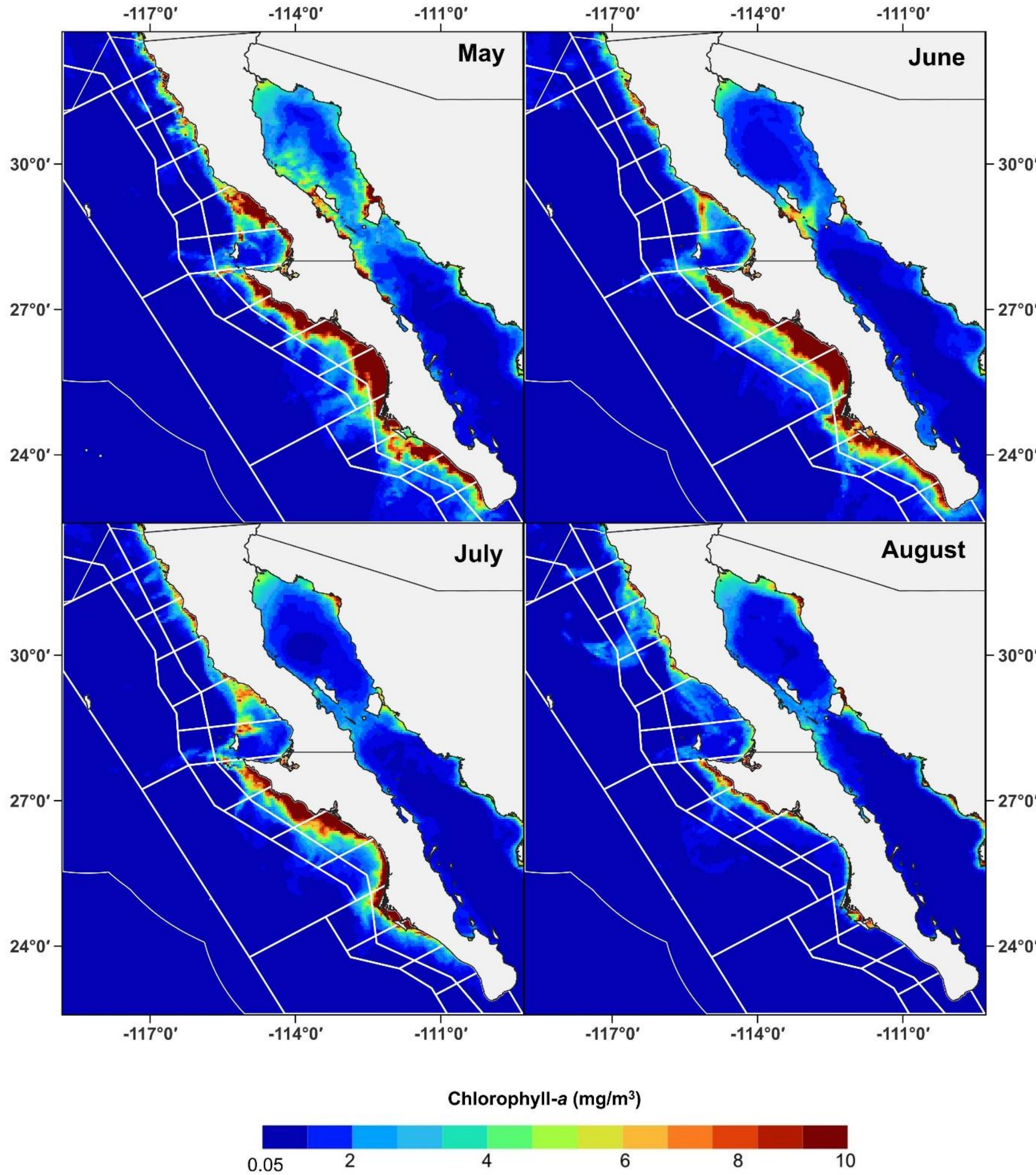

Figure 6: Satellite images of monthly CHL-a averages from May to August 2025.

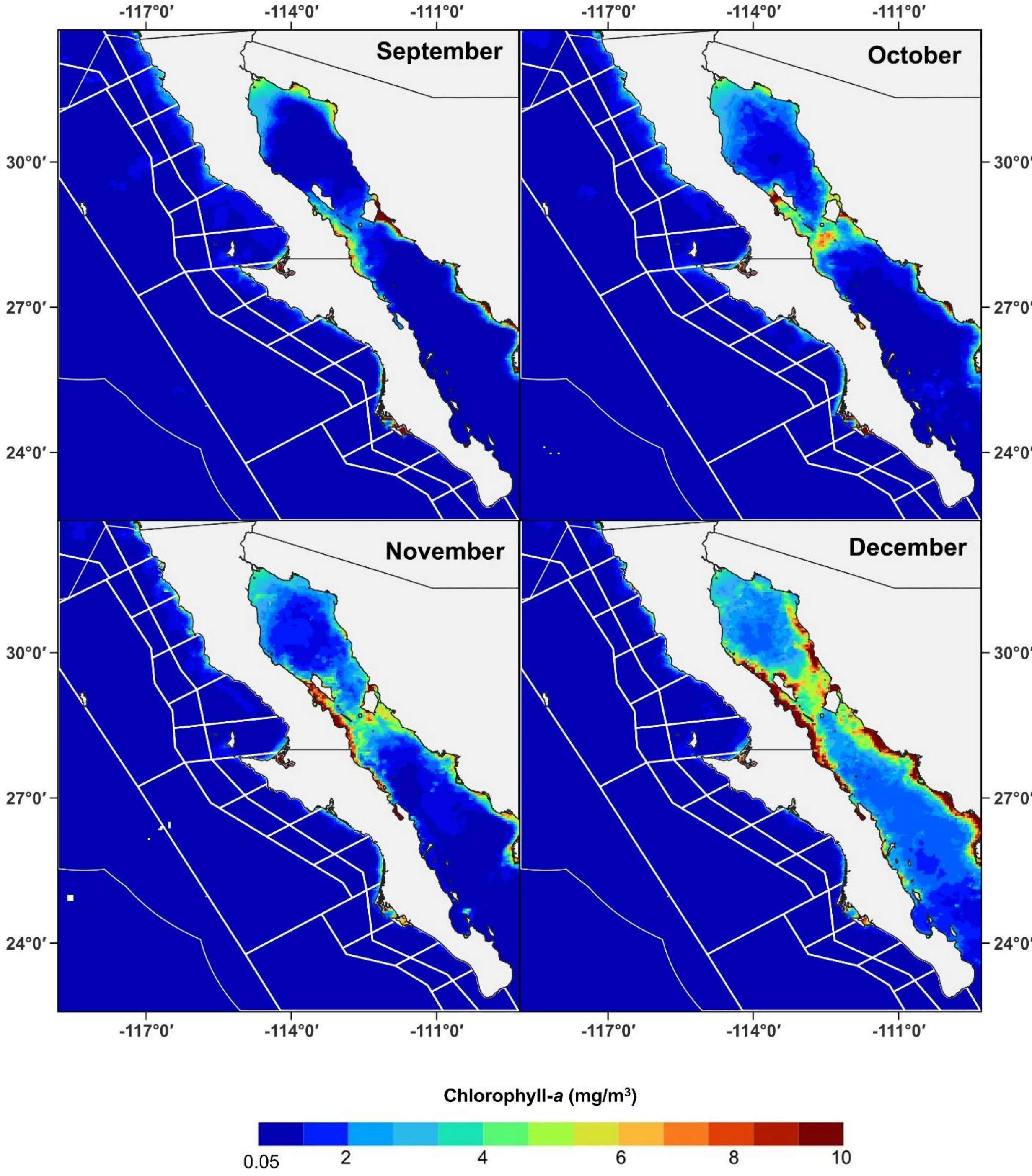

Figure 7: Satellite images of monthly CHL-a averages from September to December 2025.

Altimetry

The monthly maps depict the spatio-temporal variation of sea level anomalies (cm) and surface geostrophic current vectors around the Baja California peninsula and within the Gulf of California, showcasing a pronounced seasonal circulation regime.

From January to April, negative sea level anomalies (0 to −20 cm) predominate across both the Pacific margin and the Gulf of California. During this late winter and early spring period, negative anomalies reach their maximum extent in April throughout the Gulf, accompanied by localized mesoscale structures off the Pacific coast. The surface geostrophic velocity vectors reflect weak to moderate circulation during these early months (Figure 8).

During May and June, a transition toward positive sea level anomalies begins. Isolated anticyclonic eddies (positive SLA features) develop off the Pacific coast, while positive sea level anomalies start to enter and spread northward within the Gulf of California, where values shift from near-neutral to above 20 cm by June (Figure 9).

The period from July to September marks the annual peak in sea level anomalies. Strong positive SLA values (30 to >40 cm) dominate the entire Gulf of California and extend along the southern Pacific coast of the peninsula. These elevated values correspond to intense poleward geostrophic flows. In several inner Gulf areas during July and August, values exceed the upper detection threshold, resulting in unmapped coastal zones (represented in white) (Figure 9 and Figure 10).

In October through December, sea level anomalies progressively decline across the region. Positive anomalies recede southward from the upper Gulf in October and November, yielding to neutral and negative SLA conditions (−10 to 0 cm) throughout most of the region by December, thus returning to the winter circulation pattern (Figure 10).

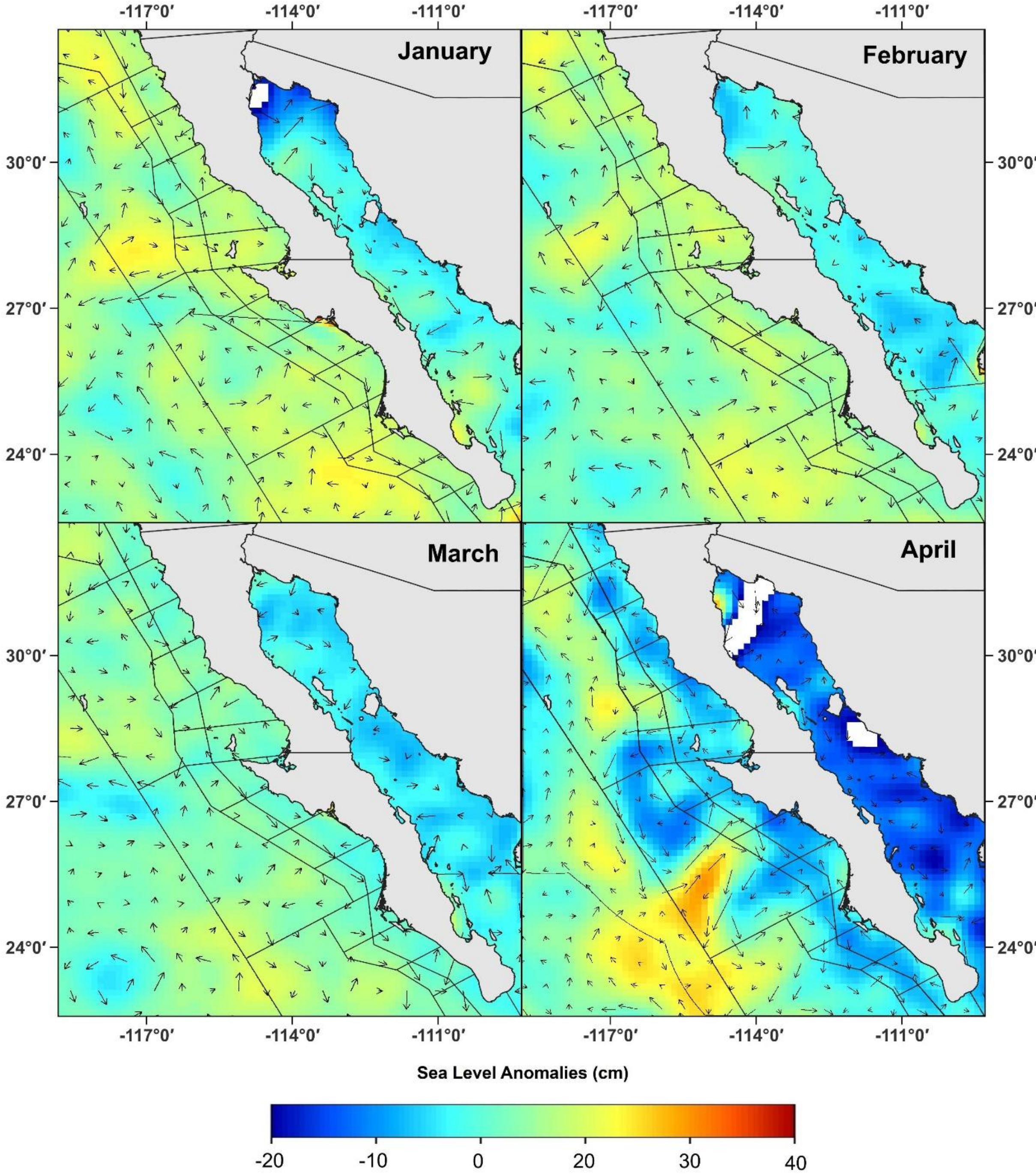

Figure 8: Satellite images of monthly SLA averages from January to April 2025.

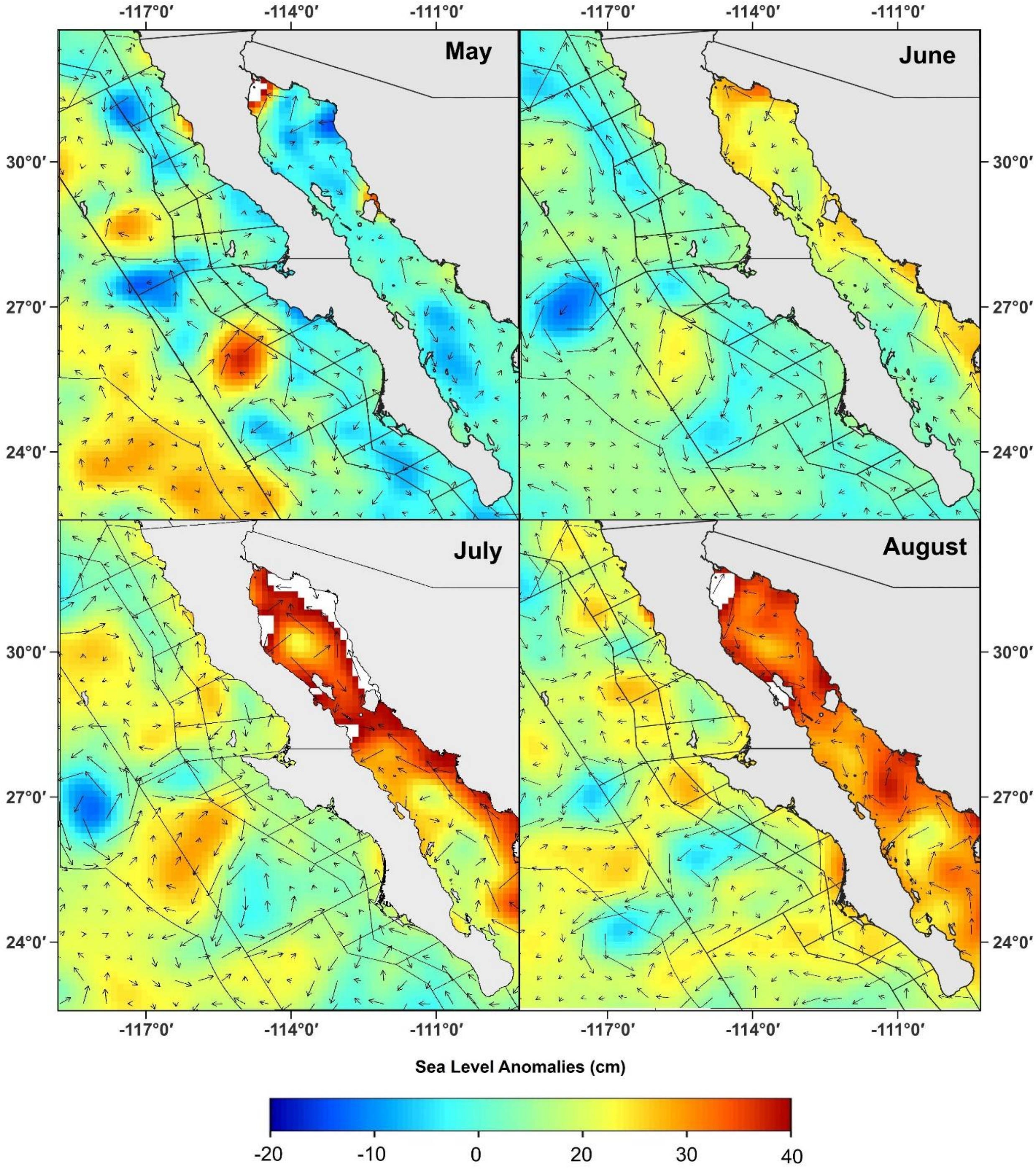

Figure 9: Satellite images of monthly SLA averages from May to August 2025.

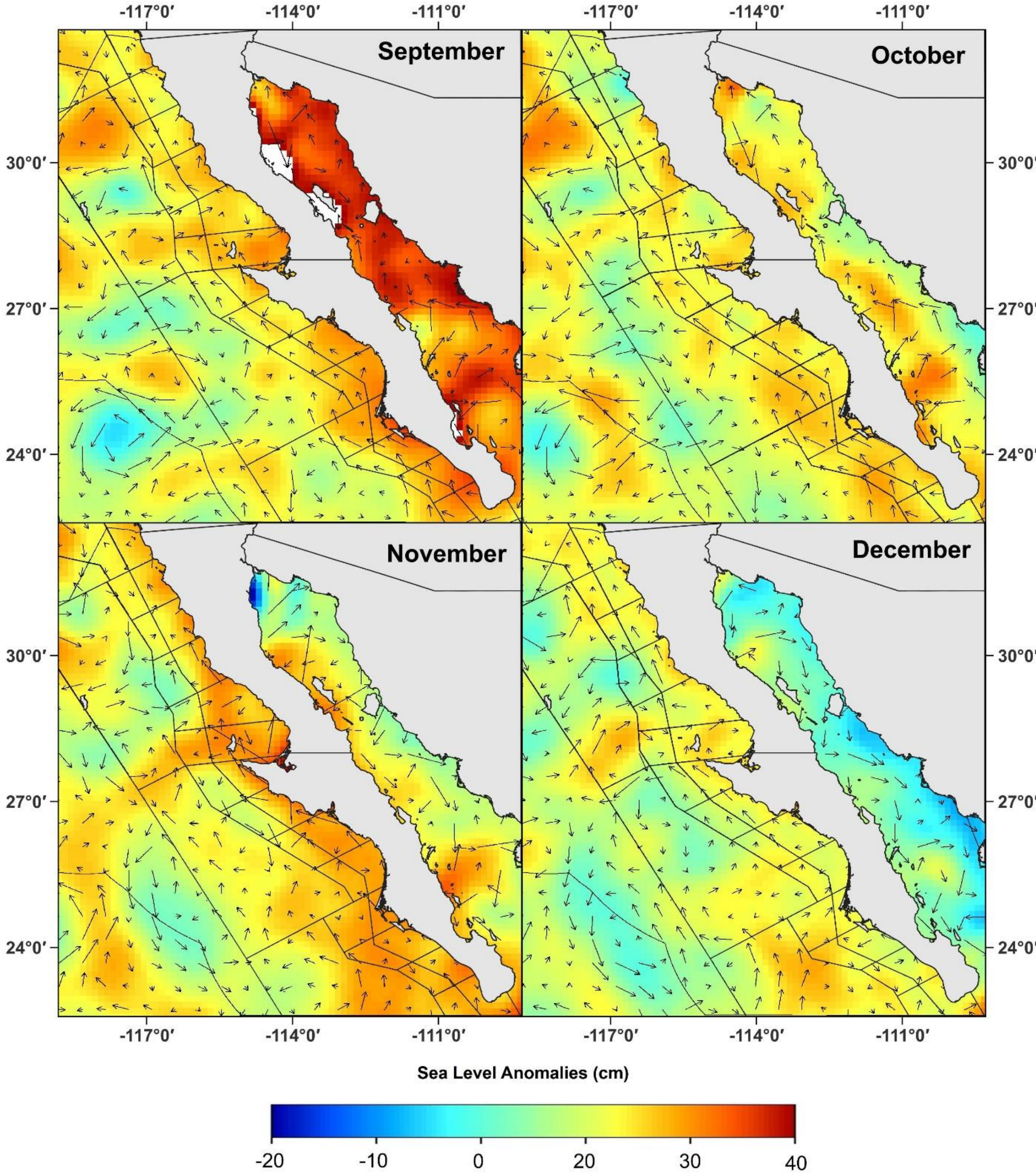

Figure 10: Satellite images of monthly SLA averages from September to December 2025.

Upwelling Indices

The northernmost station (30°N 119°W) displays relatively low and stable upwelling values throughout the year, fluctuating mostly between 30 $m^3/s/100m$ and 50 $m^3/s/100m$. It experiences moderate peaks in February and May (~48–65 $m^3/s/100m$), alongside a minor autumnal peak in October (~42 $m^3/s/100m$), before dropping to its minimum in December (~13 $m^3/s/100m$).

The central station (27°N 116°W) demonstrates sustained, moderate-to-high upwelling intensity for much of the year. After a sharp rise from January (~38 $m^3/s/100m$) to February (~75 $m^3/s/100m$), values remain consistently high throughout spring and early summer (staying above 70 $m^3/s/100m$ from February to June). A secondary peak occurs in October (~70 $m^3/s/100m$), followed by a steep decline into winter (reaching around 28–30 $m^3/s/100m$ in November and December).

The southernmost station (24°N 113°W) exhibits the most dramatic seasonal amplitude. From a low baseline in January (~38 $m^3/s/100m$), upwelling surges during spring and early summer, peaking in June at over 130 $m^3/s/100m$. This is followed by a rapid collapse through late summer, reaching a sharp minimum in September (~5 $m^3/s/100m$). A modest secondary rise occurs in October (~50 $m^3/s/100m$) before values taper off again towards the end of the year.

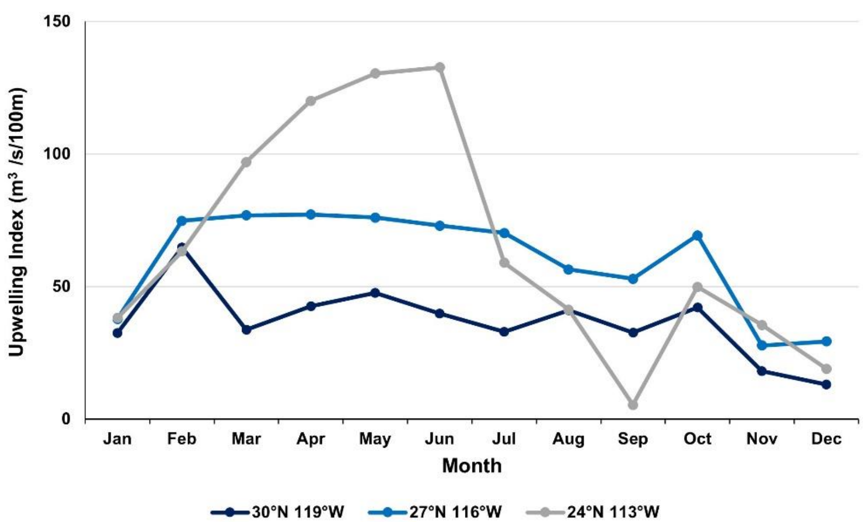

Figure 11: Monthly upwelling index values along the Pacific coast of the Baja California Peninsula at three latitudinal locations: northern (30°N 119°W), central (27°N 116°W), and southern (24°N 113°W).

Time series of Todos Santos Bay and Magdalena–Almejas Bay

In Todos Santos Bay, 2025 SSTs follow a typical seasonal cycle, ranging from a minimum of ~14.2 °C in January to a peak of ~21.9 °C in September. Values align closely with the historical mean, showing slightly cooler conditions in early winter and slightly warmer conditions from June to December, briefly touching the 22 °C threshold in late summer. In Magdalena–Almejas Bay (B1), temperatures are overall higher, ranging from ~17.2 °C in spring to a peak of ~29 °C in August. SSTs remain above the 22 °C threshold from June through November, with 2025 values exceeding historical averages during the late summer peak (Figure 12; A1 and B1).

The standardized anomaly time series reveal pronounced interannual thermal fluctuations across both bay systems. Between 2023 and early 2026, three distinct positive anomaly periods (warm phases, red shading) occur during the late summer–autumn months of each year, separated by cooler negative anomaly periods (blue shading) during winter and spring. These local SST anomaly cycles display a strong coherence with the PDO Index (dotted line), which remains predominantly negative throughout the period, and the ENSO Index (dashed line), which fluctuates around neutral to slightly positive values (Figure 12; A2 and B2).

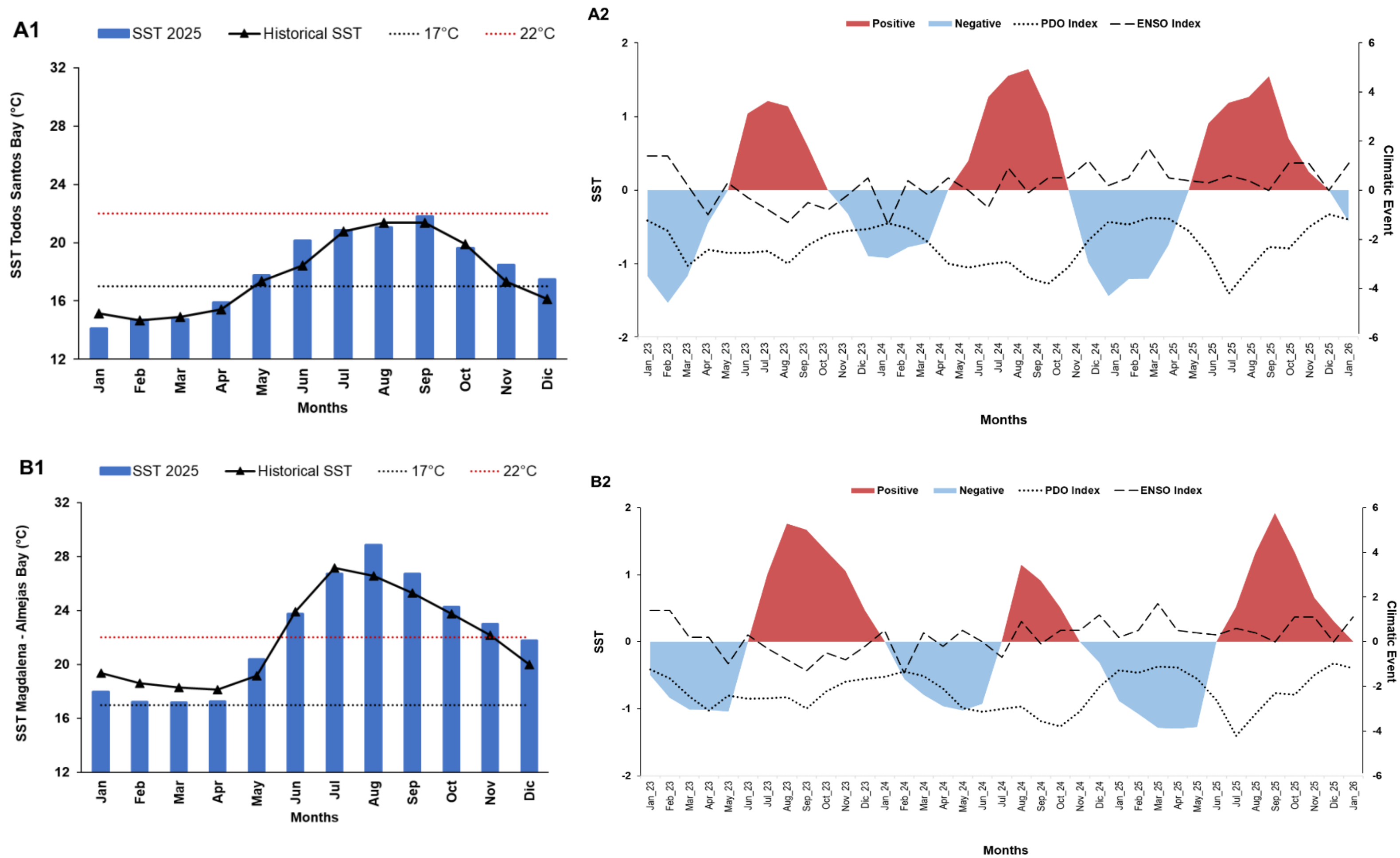

Figure 12: SST characterization for Todos Santos Bay (A1,2) and Magdalena–Almejas Bay (B1,2). (A1, B1) Monthly SST during 2025 compared with the historical mean and the isotherms (17° and 22°C). (A2, B2) Standardized SST anomaly time series from January 2023 to January 2026 showing positive (red) and negative (blue) anomaly phases, superimposed with the Pacific Decadal Oscillation Index and the El Niño–Southern Oscillation Index.

Spatio-temporal catch patterns of small pelagics in Region A

In the monthly species composition, *S. sagax* dominates total landings throughout the year. It accounts for over 50% of monthly catches in most months, reaching nearly 80% to 100% between September and December. *E. mordax* represents the second largest component, contributing significantly from spring to mid-summer (April to July), when it comprises up to 40–50% of monthly catches. Secondary species show distinct seasonal occurrences: *S. japonicus* appears primarily between February and August, *Opisthonema* spp. contributes notably in February, March, and November, and *C. mysticetus* is caught mainly in September and October. Other species (*E. teres* and *T. symmetricus*) account for minor proportions (<2%) in specific months (Figure 13).

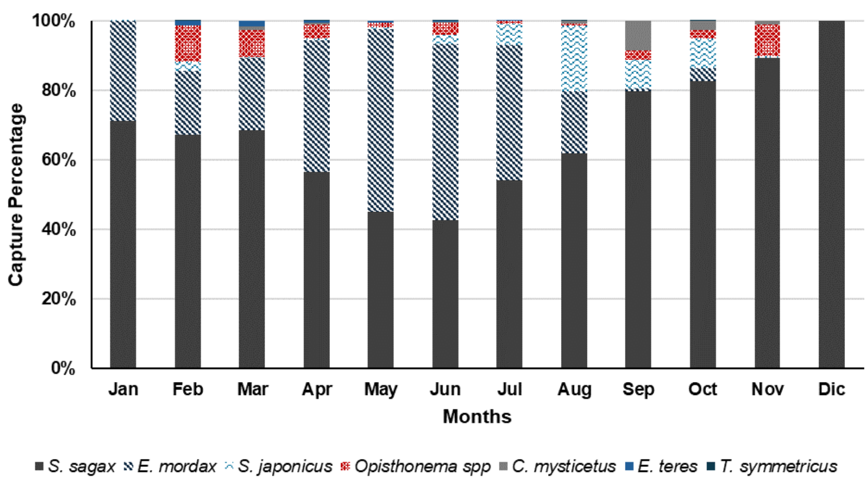

Figure 13: Monthly catch percentage of small pelagic landed in Region A during 2025.

There are regional differences in catch composition between Baja California (BC) and Baja California Sur (BCS). In BC, catches are composed almost entirely of two species: *S. sagax* (58%) and *E. mordax* (41%), with *S. japonicus* (1%) and *T. symmetricus* (0.01%) making up the remainder. Conversely, landings in BCS feature greater species diversity. While *S. sagax* remains the principal targeted species (75%), *S. japonicus* (12%), *Opisthonema* spp. (8%), and *C. mysticetus* (4%) contribute notable shares, alongside minor catches of *E. teres* (1%) and *T. symmetricus* (0.02%) (Figure 14).

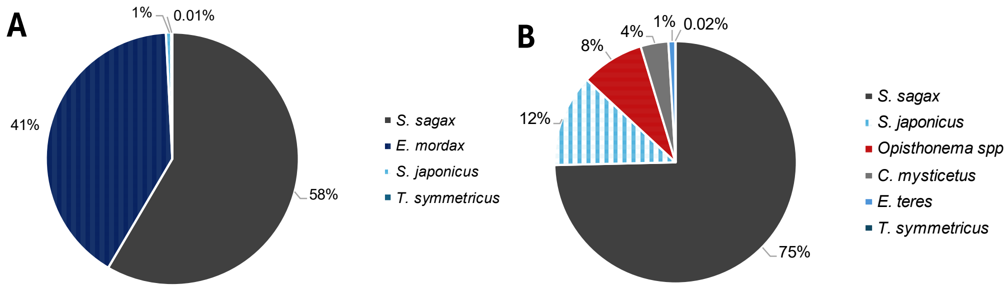

Figure 14: Species composition of small pelagic landings by state during 2025: Baja California (A) and Baja California Sur (B).

CONCLUSIONS

The biological dynamics and spatial availability of small pelagic fisheries in Region A during 2025 were directly driven by the interplay between large-scale oceanographic transport, seasonal atmospheric forcing, and local environmental variables. The physical mechanics of the California Current System establish a tight coupling between upwelling intensity, thermal structure, sea level anomalies, and primary productivity, which together determine the abundance and catchability of target species. Wind-driven coastal upwelling serves as the primary catalyst for ecosystem productivity in the region. High Bakun upwelling indices during late winter and spring drive the Ekman transport of cold, nutrient-rich deep waters to the surface. This process directly triggers phytoplankton blooms, as reflected in elevated chlorophyll-*a* concentrations (>10 mg/m^3). These nutrient-dense, high-productivity conditions coincide with cool sea surface temperatures (14–18 °C) and negative sea level anomalies (−20 to 0 cm). Temperate small pelagics, particularly Northern anchovy (*E. mordax*) and Pacific sardine (*S. sagax*), feed and aggregate within these upwelling plumes, resulting in substantial landings of anchovy (comprising up to 50% of monthly catches) during the spring-to-early-summer upwelling maximum. Conversely, during late summer and autumn, the suppression of upwelling — most notably at the southern station (24°N), where indices drop to ~5 m^3/s/100m — leads to a dramatic shift in environmental conditions. The influx of warm tropical and subtropical water masses causes sea surface temperatures to peak (reaching 24–

26 °C along the Pacific coast and >28 °C in southern bays) and drives strong positive sea level anomalies (30 to >40 cm) associated with poleward geostrophic flow. Under these oligotrophic, thermally stratified conditions, chlorophyll-*a* levels collapse across open waters. Temperate species such as *E. mordax* retreat or diminish in catchability, whereas warm-affinity subtropical species including thread herring (*Opisthonema* spp.), Northern anchoveta (*C. mysticetus*), and Pacific chub mackerel (*S. japonicus*) become accessible to the southern fleet in Baja California Sur. Furthermore, *S. sagax* shifts to absolute dominance (80–100% of catches) in late autumn as the system begins its transition back to winter cooling. Standardized anomaly time series demonstrate that local thermal peaks mirror broader climatic oscillations such as ENSO and PDO, highlighting that systematic monitoring of these coupled oceanographic variables is essential for anticipating stock movements and implementing adaptive management strategies for small pelagic fisheries.

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